Square-class patterns of seven consecutive integers and seven elementary integral quartics

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Abstract

We study the affine rank of the square classes of seven consecutive nonzero integers. This records when a common rational scaling makes every term a square in a field of degree at most four generated by two square roots. After affine rank at most one is excluded, enumeration up to relabelling gives 651 three- or four-block equality patterns; quotienting by reversal and applying known six-term and four-square restrictions leaves 284 candidates, each controlled by precisely fifteen character quotients. A four-consecutive-factor identity eliminates 186 patterns. We determine all integral points on $y^2 = t(t+2)(t+3)(t+6)$ by an elementary squarefree-kernel argument and finite congruence certificates. Together with an integral reflection, this eliminates 44 further patterns after a same-parameter check, leaving 54. Two constant-difference factorizations exclude another 19 and 12 patterns. A third such factorization excludes eight more. Two integral translates and a final constant-difference factorization exclude five, three and three further patterns, respectively, leaving four explicitly listed necessary candidates. Every computational count and congruence certificate is bound by a versioned reproducibility manifest. No bounded search or unproved Mordell–Weil computation enters the proof. The theorem neither proves that any remaining pattern is realizable nor decides the unrestricted rational-progression problem.

Keywords: square classes; arithmetic progressions; integral points; biquadratic fields; Diophantine equations.

Mathematics Subject Classification (2020): Primary 11B25; Secondary 11D45, 11G05.

1 Square-class formulation

Put $G = \mathbf{Q}^*/\mathbf{Q}^{*2}$, written additively over $\mathbf{F}_2$. For the nonzero numbers $a_i = t + i$, define

$$\operatorname{arank}(a_0, \ldots, a_6) = \dim_{\mathbf{F}_2} \langle [a_i] - [a_0] : 1 \le i \le 6 \rangle.$$

The parameters $t = -6, \ldots, 0$ are excluded.

For reference, define

$$R_s(N) = \max_{\substack{a,q \in \mathbf{Q},\ q \neq 0;\ V \le G,\ \dim_{\mathbf{F}_2} V = s \\ a+qi \neq 0\ (0 \le i < N)}} \#\{0 \le i < N : [a + qi] \in V\},$$

so every squareclass occurring in the definition is defined. Thus the maximum ranges over nonconstant rational arithmetic progressions whose N displayed terms are all nonzero, and over exactly s-dimensional $\mathbf{F}_2$-subspaces of G. The maximum is unchanged if one permits a common rational scaling and replaces linear rank by affine rank.

Lemma 1. $R_1(6) = 5$.

Proof. If six rational terms have classes in $V = \langle [D] \rangle$, they are six squares in the quadratic field $\mathbf{Q}(\sqrt{D})$. Xarles proves that a nonconstant progression over a quadratic field has at most five square terms [13, Section 4]. Conversely,

$$49, 169, 289, 409, 529$$

is a nonconstant progression of squares over $\mathbf{Q}(\sqrt{409})$; appending the sixth term 649 shows that five of six terms meet the condition, so the upper bound is attained. $\square$

Proposition 2 (Common scaling)**.** *The seven classes have affine rank at most two if and only if there are a common $\lambda \in \mathbf{Q}^*$ and a field $L = \mathbf{Q}(\sqrt{D_1}, \sqrt{D_2})$ of degree at most four such that every $\lambda(t+i)$ is a square in L. If the difference classes span a two-dimensional space, L may be chosen biquadratic.*

Proof. Write $[t+i] = c + v_i$ with $v_i \in V = \langle [D_1], [D_2] \rangle$, and choose $[\lambda] = c$. Then $[\lambda(t+i)] = v_i$ dies in L^*/L^{*2}. Conversely,

$$\ker\bigl(\mathbf{Q}^*/\mathbf{Q}^{*2} \longrightarrow L^*/L^{*2}\bigr) = \langle [D_1], [D_2] \rangle,$$

so the classes lie in a common affine coset of dimension at most two. Indeed, if a rational class $[r]$ dies in L, then either r is already a rational square or $\mathbf{Q}(\sqrt{r})$ is a quadratic subfield of L. The quadratic subfields of a biquadratic extension are exactly $\mathbf{Q}(\sqrt{D_1})$, $\mathbf{Q}(\sqrt{D_2})$, and $\mathbf{Q}(\sqrt{D_1 D_2})$, by the Kummer correspondence. This gives the stated kernel; if $[L : \mathbf{Q}] < 4$, the same argument gives the corresponding zero- or one-dimensional subspace. $\square$

Remark 3. *The scaling is essential: the common class c can be independent of the two difference classes, in which case adjoining $\sqrt{c}$ may raise the degree to eight. We do not claim the unscaled $t+i$ are squares in one fixed biquadratic field.*

2 The finite pattern space

Represent a rank-two configuration by labels $\ell_i \in \mathbf{F}_2^2$. Lemma 1 forces both endpoint six-term windows to contain at least three labels. A block of size at least five is excluded by $Q(7) = 4$ [7, Corollary 6]. A block at four positions (i, j, k, l) with $j - i = l - k$ is a symmetric four-subset of $\{0, \ldots, 6\}$; after translation and division by the common gap it is one of the five exceptional sets in [7, Proposition 5], and is therefore excluded. Indeed its positive successive gaps are (a, b, a) with $2a + b \leq 6$; after division by $\gcd(a, b)$ the possibilities are $(1,1), (1,2), (1,3), (1,4), (2,1)$.

Proposition 4 (Exact pattern count)**.** *Before these screens there are 343 label patterns up to reversal. The screens eliminate 59 reversal orbits, leaving 284.*

Proof. Since $\mathrm{AGL}(2,2) \cong S_4$, an orbit using three or four labels is an unlabelled partition into three or four blocks. Restricted growth strings enumerate

$$S(7,3) + S(7,4) = 301 + 350 = 651$$

partitions. Reversal fixes 35, so Burnside gives $(651 + 35)/2 = 343$.

The finite screen discards a word if an endpoint six-window has at most two blocks, if a block has size at least five, or if a four-element block has equal first and last gaps. It discards 109 words forming 59 reversal orbits. Of the remaining 542 words, 26 are reversal-fixed, giving $(542 + 26)/2 = 284$. The supplementary generator directly asserts every count. $\square$

For a label word define

$$T : \mathbf{F}_2^7 \longrightarrow \mathbf{F}_2 \oplus \mathbf{F}_2^2, \qquad e \longmapsto \left(\sum e_i, \sum e_i \ell_i\right).$$

Lemma 5 (Fifteen characters)**.** *For each pattern,* $\ker T$ *has dimension four and exactly 15 nonzero elements. Every such* e *gives the necessary equation*

$$y_e^2 = \prod_{e_i=1} (t+i).$$

Proof. The labels affinely span $\mathbf{F}_2^2$, so T has rank three. Rank–nullity gives dimension four. If $[t+i] = c + \phi(\ell_i)$, the class of the product is $(\sum e_i)c + \phi(\sum e_i \ell_i) = 0$. These are all affine linear relations. ☐

3 The first 186 exclusions

Lemma 6 (Four consecutive factors)**.** *If* $x \in \mathbf{Z}$ *and* $x(x+1)(x+2)(x+3)$ *is a rational square, then* $x \in \{-3, -2, -1, 0\}$.

Proof. It is an integer square y^2. Put $n = x(x+3)$. Then

$$x(x+1)(x+2)(x+3) = (n+1)^2 - 1,$$

so $(n+1-y)(n+1+y) = 1$. The two integral factor pairs give $n = 0$ or -2, and the four asserted values follow. ☐

The consecutive masks are $15, 30, 60, 120$. Recomputing all 15-element kernels and intersecting with these masks eliminates exactly 186 of 284 patterns, leaving 98. This is finite exact computation, not height search.

4 The mask 77 curve

Consider

$$H: \quad y^2 = t(t+2)(t+3)(t+6).$$

It is birational to $E : Y^2 = X^3 - 36X$ by

$$X = 12 + \frac{36}{t}, \quad Y = \frac{36y}{t^2}, \qquad t = \frac{36}{X-12}, \quad y = \frac{36Y}{(X-12)^2}.$$

The boundary correspondence is

H	E
$(0,0)$	O
$(-2,0)$	$(-6,0)$
$(-3,0)$	$(0,0)$
$(-6,0)$	$(6,0)$
∞_+	$(12,36)$
∞_-	$(12,-36)$.

This is Cremona 576h2, equivalently LMFDB 576.c3 [9]. The label is only a cross-check: the condition $36/(X-12) \in \mathbf{Z}$ is not ordinary integrality of X.

Theorem 7.

$$H(\mathbf{Z}) = \{(-6,0),(-3,0),(-2,0),(0,0),(6,72),(6,-72)\}.$$

Proof. Put $A = t(t+6)$ and $B = (t+2)(t+3)$. Since $B-A = 6-t$ and $A = (t+12)(t-6)+72$, one has $\gcd(A,B) \mid 72$. At a nonbranch integral point, $B > 0$ and $AB = y^2 > 0$, hence $A > 0$. Their common positive squarefree kernel is

$$d \in \{1,2,3,6\}, \qquad A = dU^2, \quad B = dV^2.$$

With $x = t+3$ this becomes

$$x^2 - dU^2 = 9, \qquad x(x-1) = dV^2.$$

Write $d = ab$ as an ordered coprime factorization. For $x > 0$ and $x < 0$ one obtains, respectively,

$$\begin{aligned} &x = aR^2,\ x-1 = bS^2, && aR^2 - bS^2 = 1,\\ &x = -aR^2,\ x-1 = -bS^2, && bS^2 - aR^2 = 1, \end{aligned}$$

together with $a^2R^4 - dU^2 = 9$. There are 18 branches. Direct exhaustion of (R,S,U) modulo the listed modulus eliminates 15:

d	$(a,b), \operatorname{sign}(x)$	modulus
1	$(1,1),+;\ (1,1),-$	16; 8
2	$(1,2),-;\ (2,1),+;\ (2,1),-$	9; 8; 8
3	$(1,3),+;\ (1,3),-;\ (3,1),+$	9; 8; 8
6	$(1,6),+;\ (1,6),-;\ (2,3),+;\ (2,3),-$	9; 8; 8; 8
	$(3,2),-;\ (6,1),+;\ (6,1),-$	8; 8; 8.

For each row the certificate records the branch equations and the modulus; the accompanying generator and regression test exhaust every residue triple. Thus each row is a finite proof rather than a bounded search.

For $d = 2, (a,b) = (1,2), x > 0$,

$$R^2 - 2S^2 = 1, \quad R^4 - 2U^2 = 9, \quad (U - RS)(U + RS) = (R^2 - 9)/2.$$

$R = 1$ is impossible, $R = 3$ gives $(S,U) = (2,6)$, and for $R \geq 5$ both factors are positive while $S > R/2$ makes the second factor larger than their product.

For $d = 3, (a,b) = (3,1), x < 0$,

$$S^2 - 3R^2 = 1, \quad U^2 = 3(R^4 - 1), \quad (RS - U)(RS + U) = R^2 + 3.$$

$R = 0, 2$ are impossible, $R = 1$ gives $(2,0)$, and for $R \geq 3$, $RS > \sqrt{3}R^2 > R^2 + 3$.

For $d = 6, (a,b) = (3,2), x > 0$,

$$3R^2 - 2S^2 = 1, \quad 3R^4 - 2U^2 = 3, \quad (U - RS)(U + RS) = (R^2 - 3)/2.$$

R is odd; $R = 1$ gives $(1,0)$, and the same factor-size contradiction holds for $R \geq 3$. These branches yield $t = 6, y = \pm 72$ and the separated branch points $t = -6, 0$. Adding all four roots proves the theorem. $\square$

5 Same-parameter elimination

Mask 77 has support $\{0,2,3,6\}$ and mask 89 has support $\{0,3,4,6\}$. The integral reflection $t=-u-6$ gives

$$t(t+3)(t+4)(t+6)=u(u+2)(u+3)(u+6).$$

Thus the nonbranch parameter sets are $\{6\}$ and $\{-12\}$.

Theorem 8 (Finite classification)**.** *Among the 98 patterns left above, 26 contain mask 77, 25 contain mask 89, and 7 contain both. All 44 patterns in the union fail the simultaneous 15-character conditions. Exactly 54 candidate patterns remain.*

Proof. The union has $26+25-7=44$ elements. For a pattern containing only mask 77, substitute $t=6$ into all 15 character products; at least one other product is nonsquare. For mask 89 use $t=-12$. If both occur, the singleton candidate sets are disjoint. The certificate records every failed mask, every common candidate set, and all 54 surviving identifiers. It independently asserts 15 distinct masks in every input row. □

6 A fourth-round gate: mask 102

The remaining 54 patterns were ranked by two exact quantities: the number of patterns containing a given four-root character, and whether its four linear factors admit a pairing into two monic quadratics with constant difference. Mask 102, supported at $\{1,2,5,6\}$, ranks first: it occurs in 19 patterns and has

$$A=(t+1)(t+6),\qquad B=(t+2)(t+5),\qquad B-A=4.$$

Proposition 9. *The integral points on*

$$y^2=(t+1)(t+2)(t+5)(t+6)$$

are exactly

$$(-6,0),\quad (-5,0),\quad (-2,0),\quad (-1,0).$$

Proof. For $t\le -7$ or $t\ge 0$, both A and B are positive. Moreover $\gcd(A,B)\mid 4$. If AB is a square, the common positive squarefree kernel is $d=1$ or 2, so $A=dU^2, B=dV^2$. For $d=1$,

$$(V-U)(V+U)=V^2-U^2=4.$$

The only positive factor pair of equal parity is $(2,2)$, forcing $U=0$ and hence a branch point. For $d=2$ one obtains $V^2-U^2=2$, impossible modulo 4. The six middle integers $-6\le t\le -1$ give right sides $0,0,12,12,0,0$, respectively. This proves the list. □

Corollary 10. *Mask 102 eliminates 19 of the 54 patterns in Theorem 8. Hence 35 candidate patterns remain after this fourth-round gate.*

Proof. All integral points of Proposition 9 have $t\in\{-6,\dots,0\}$ and are therefore degenerate for the original seven-term problem. The exact occurrence table contains mask 102 in 19 of the 54 rows. Thus no same-parameter candidate survives in those rows. □

7 A fifth-round gate: mask 108

Mask 108 has support $\{2, 3, 5, 6\}$ and occurs in 12 of the 35 patterns.

Proposition 11. *The integral points on*

$$y^2 = (t+2)(t+3)(t+5)(t+6)$$

are exactly

$$(-6,0), (-5,0), (-4,-2), (-4,2), (-3,0), (-2,0).$$

Proof. Put $A = (t+2)(t+6)$ and $B = (t+3)(t+5) = A+3$. For $t \le -7$ or $t \ge -1$, both are positive and $\gcd(A,B) \mid 3$. If AB is a square, their common positive squarefree kernel is $d = 1$ or 3, so $A = dU^2, B = dV^2$. For $d = 1$, $(V-U)(V+U) = 3$ gives $(U,V) = (1,2)$, while $A = 1$ would give $(t+4)^2 = 5$, impossible modulo 8. For $d = 3$, $(V-U)(V+U) = 1$ forces $U = 0$, contradicting $A > 0$. At the five middle integers $-6 \le t \le -2$, direct substitution gives the displayed six points. □

8 A sixth-round gate: mask 99

Mask 99 has support $\{0, 1, 5, 6\}$ and occurs in eight of the 23 patterns.

Proposition 12. *The integral points on*

$$y^2 = t(t+1)(t+5)(t+6)$$

are exactly

$$(-6,0), (-5,0), (-3,-6), (-3,6), (-1,0), (0,0).$$

Proof. Put $A = t(t+6)$ and $B = (t+1)(t+5) = A+5$. For $t \le -7$ or $t \ge 1$, both are positive and $\gcd(A,B) \mid 5$. If AB is a square, their common positive squarefree kernel is $d = 1$ or 5, so $A = dU^2, B = dV^2$. For $d = 1$, $(V-U)(V+U) = 5$, hence $(U,V) = (2,3)$; then $A = 4$ gives $(t+3)^2 = 13$, impossible modulo 8. For $d = 5$, $(V-U)(V+U) = 1$ forces $U = 0$, contradicting $A > 0$. Direct substitution for $-6 \le t \le 0$ gives the displayed points. □

9 A seventh-round gate: mask 51

Mask 51 has support $\{0, 1, 4, 5\}$ and occurs in five of the 15 patterns.

Proposition 13. *The integral points on*

$$y^2 = t(t+1)(t+4)(t+5)$$

are exactly

$$(-5,0), (-4,0), (-1,0), (0,0).$$

Proof. The substitution $s = t-1$ identifies the equation with the mask-102 equation of Proposition 9, and is a bijection on integral parameters. For an independent elementary check, put $A = t(t+5)$ and $B = (t+1)(t+4) = A+4$. For $t \le -6$ or $t \ge 1$, both are positive and $\gcd(A,B) \mid 4$. Their common positive squarefree kernel is $d = 1$ or 2. If $d = 1$, then $(V-U)(V+U) = 4$ forces $U = 0$, contrary to $A > 0$; if $d = 2$, then $V^2 - U^2 = 2$, impossible modulo 4. Direct substitution for $-5 \le t \le 0$ gives right sides $0, 0, 12, 12, 0, 0$. □

10 An eighth-round gate: mask 90

Mask 90 has support $\{1,3,4,6\}$ and occurs in three of the 10 patterns.

Proposition 14. *The integral points on*

$$y^2 = (t+1)(t+3)(t+4)(t+6)$$

are exactly

$$(-6,0), (-4,0), (-3,0), (-1,0).$$

Proof. Since the right side is an integer, a rational value of y is automatically an integer. Put

$$A = (t+1)(t+6), \qquad B = (t+3)(t+4) = A+6.$$

For $t \le -7$ or $t \ge 0$, both are positive and $\gcd(A,B) = \gcd(A,6)$. If AB is a square, their common positive squarefree kernel is $d \in \{1,2,3,6\}$, so $A = dU^2$, $B = dV^2$ and

$$d(V^2 - U^2) = 6.$$

For $d = 1$ and $d = 3$, the resulting differences of two squares are respectively 6 and 2, both impossible modulo 4. For $d = 2$, $(V-U)(V+U) = 3$ forces $(U,V) = (1,2)$ and $A = 2$; but then $(2t+7)^2 = 4A + 25 = 33$, impossible since $5^2 < 33 < 6^2$. For $d = 6$, $(V-U)(V+U) = 1$ forces $U = 0$, contrary to $A > 0$. At the six remaining integers $-6 \le t \le -1$, the right sides are $0, -8, 0, 0, -8, 0$, giving precisely the displayed points. □

11 A ninth-round gate: mask 54

Mask 54 has support $\{1,2,4,5\}$ and occurs in three of the seven patterns left by Proposition 14.

Proposition 15. *The integral points on*

$$y^2 = (t+1)(t+2)(t+4)(t+5)$$

are exactly

$$(-5,0), (-4,0), (-3,-2), (-3,2), (-2,0), (-1,0).$$

Proof. The substitution $s = t-1$, with $Y = y$, gives the factorwise identity

$$(s+2)(s+3)(s+5)(s+6) = (t+1)(t+2)(t+4)(t+5).$$

It is a bijection on integral parameters, with inverse $t = s+1$, so the result also follows point by point from Proposition 11.

For an independent complete check, put

$$A = (t+1)(t+5), \qquad B = (t+2)(t+4) = A+3.$$

For $t \le -6$ or $t \ge 0$, both are positive and $\gcd(A,B) = \gcd(A,3)$. If AB is a square, their common positive squarefree kernel is $d \in \{1,3\}$, so $A = dU^2$, $B = dV^2$ and $d(V^2 - U^2) = 3$. For $d = 1$, the only positive factorization $(V-U)(V+U) = 3$ gives $(U,V) = (1,2)$ and $A = 1$; then $(t+3)^2 = A + 4 = 5$, impossible modulo 8. For $d = 3$, $(V-U)(V+U) = 1$ gives $U = 0$, contrary to $A > 0$. At the five remaining integers $-5 \le t \le -1$, the right sides are $0, 0, 4, 0, 0$, giving precisely the displayed points. As before, rational y is automatically integral. □

Theorem 16 (Summary classification). *Let $t \in \mathbf{Z} \setminus \{-6, -5, \ldots, 0\}$. If the seven nonzero classes $[t], \ldots, [t+6]$ have affine rank at most two, then their equality partition, modulo relabelling and reversal, is one of the four patterns in Table 1. In particular, the exact reduction chain is*

$$651 \longrightarrow 343 \longrightarrow 284 \longrightarrow 98 \longrightarrow 54 \longrightarrow 35 \longrightarrow 23 \longrightarrow 15 \longrightarrow 10 \longrightarrow 7 \longrightarrow 4.$$

This theorem neither asserts that any of the four patterns is realizable nor decides $R_2(7)$.

Proof. Affine rank zero or one is excluded by Lemma 1, after applying the common rational scaling, to an endpoint window. Thus the actual affine rank is two. The map from the affinely spanning labels to squareclasses therefore has rank two and is injective, so the labels are precisely the actual squareclass equality blocks. Proposition 4 and Lemma 5 give 284 self-contained packets. Lemma 6 leaves 98; Theorem 8 leaves 54; Proposition 9 leaves 35; and Proposition 11, whose points all have a zero among the seven original terms, eliminates 12 more. Proposition 12 eliminates eight of those 23 rows, since all its integral points likewise have a zero among the original seven terms. The integral points of Proposition 13 are also degenerate and its mask occurs in five of the 15 remaining rows. Finally, Proposition 14 has only degenerate points, and mask 90 occurs in three of the 10 remaining rows. Proposition 15 has only points with a zero among the original seven terms, and mask 54 occurs in three of those seven rows. The remaining exact occurrence audit gives Table 1. □

Table 1: Final canonical candidates after the mask-54 gate. ID is the zero-based index in the supplement's `unresolved_patterns_ranked`, ordered by its documented difficulty score and then by restricted-growth word.

ID	partition	ID	partition	ID	partition
12	0012202	31	0001202	134	0012131
276	0010203				

12 Supplement and reproducibility

The accompanying supplement is identified by release `paper-square-supplement-v0.9.0`. Its root manifest is `PAPER_SQUARE_SUPPLEMENT_MANIFEST.json`; it records SHA-256 and byte length for every generator, input/output certificate and regression test, together with Python/SymPy versions and clean reproduction commands. The manifest SHA-256 is the concatenation of the following two lines:

```
e218f10e116ec7732c9d369384bc0615
6195ea4cc62165cd3107583b03546c6d
```

That manifest retains its explicit local-candidate status. The current public working tree, including later audits and submission prose, is at `https://github.com/lixiang90/vibemath`. Its project directory is the concatenation

```
square-progressions/
seven-consecutive-squareclasses
```

and is bound by the repository-root `MANIFEST.sha256`. Before journal submission the authors must freeze the exact payload under an immutable release URL or preservation DOI and verify the downloaded bytes. A rebuilt PDF is required to agree in mathematical content and diagnostics, not necessarily byte for byte, because PDF metadata timestamps may vary.

The full regression command is stored verbatim in the manifest. The finite congruence certificate stores, for each of the 15 excluded mask-77 branches, its equations and obstruction modulus; its generator and regression test implement and exhaust the corresponding residue predicate. The remaining three branches carry symbolic factor identities and thresholds.

Online Resource 1 (supplementary code and certificates). The supplementary ZIP contains the exact Python generators, regression tests, JSON certificates and nested manifest used to audit the finite pattern counts and the mask-77, mask-89, mask-102, mask-108, mask-99, mask-51, mask-90 and mask-54 exclusions. It contains no additional mathematical claim beyond the theorem and claim boundary stated in this article. Before submission, the file metadata and upload form must add the approved article title, journal name, author names, affiliations and corresponding-author e-mail address.

13 Prior-art boundary and limitations

Work over number fields [13, 6, 1, 4, 5] asks for terms that are themselves squares in a prescribed extension. Our affine formulation allows Proposition 2's common rational scaling. Nearby algorithmic work concerns S-integral points [11], simultaneous Pell equations [10], and integral points in progression on congruent-number curves [2]. There is also a substantial literature on square products of selected terms or blocks in progressions and on the associated hyperelliptic curves over short intervals [12, 8, 3]. Those works do not, as far as our search found, state the affine-rank-two equality-pattern classification or carry out the simultaneous 15-character, same-parameter audit used here. We did not locate the combined seven-term reduction of Theorem 16. This is a *not found* report, not proof of novelty; in particular, the elementary quartic point lists are not claimed to be independently new.

The manifest and nested certificates bind SAFE and Round-04 inputs by SHA-256 and store the 18 branches, structured factor identities, 44 same-parameter audits, and 54 remaining identifiers. The fourth-round certificate separately stores the mask-102 ranking, proof data, the 19 excluded identifiers and the 35 intermediate survivors; the mask-108 certificate stores its complete point proof data and 12 exclusions; the mask-99 certificate records its full 23-pattern occurrence inventory, complete point proof, eight exclusions and 15 intermediate candidates; the mask-51 certificate records the complete genus-one inventory, point proof, five exclusions and 10 intermediate candidates; the mask-90 certificate records all four squarefree-kernel branches, its complete point proof, three exclusions and seven intermediate candidates; and the mask-54 certificate records its integer-point bijection with mask 108, independent two-kernel proof, three exclusions and four final candidates. An end-to-end test regenerates 651–343–284 and all $284 \cdot 15 = 4260$ characters. We do not prove that the final four patterns are realized, or that no seven-term rank-two progression exists. The 2026 papers have additional rank and class-number hypotheses; none is inferred from a label here.

Statements and Declarations

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Use of large language models and AI-assisted tools. Large language models were used in this project for mathematical exploration, generation and revision of computer code, literature-search assistance, and generation and revision of manuscript and submission text. No AI system is an author. Before submission, the human authors must independently verify every proof, computation, code path, citation and statement, confirm that this disclosure accurately describes the use made, approve the final text and accept full responsibility for its integrity.

Data and code availability. The code, certificates, audits and submission materials are publicly available in the working repository at `https://github.com/lixiang90/vibemath`, under `square-progressions/seven-consecutive-squareclasses`. **AUTHOR CONFIRMATION REQUIRED.** Before submission, the human authors must approve an immutable release, insert its genuine persistent identifier, and verify the downloaded archive against the published SHA-256 manifest. No preservation DOI is claimed in this draft.

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